

Integrating High-Level Visual Features for YouTube View Forecasting: A Neural Network-Based Approach

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Abstract

This research presents a novel framework for forecasting YouTube video viewership by integrating state-of-the-art action feature extraction with two advanced semantic analysis techniques: a pre-trained transformer encoder and a Bag-of-Words encoder. A custom-designed neural network predictor facilitates the fusion of semantic and temporal information, yielding an enhanced predictive architecture. Based on sophisticated regression methodologies, this approach establishes a new benchmark in video analytics and viewer engagement prediction. Beyond its theoretical contributions, the proposed method offers practical implications for optimizing digital content strategies, representing a significant advancement in multimedia analytics.

Keywords

Video Popularity Prediction, Multimodal Learning, User Engagement Prediction

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1 Introduction

The ascendance of online video consumption is a defining trend of our digital age, with the average viewer now dedicating approximately 17 hours per week to digital video content as of 2023. This significant engagement is further highlighted by the statistic that 92% of global internet users engage with digital video content on a weekly basis, as reported by Oberlo.com in 2024[4].

Reflecting the growing economic importance of this trend, the market for online video platforms was valued at 8.09 billion in 2022, with projections suggesting a compound annual growth rate (CAGR) of 20.7% from 2023 to 2030 [5]. This market's remarkable growth trajectory can be attributed to the increasing consumer preference for online video content over traditional mediums, such as cable and satellite television, as detailed by Grand View Research in 2023 [5].

YouTube, a bellwether in the online video domain, reported a milestone of approximately 100 million paid subscribers across its

Music and Premium services as of February 2024, illustrating a noteworthy expansion from the 50 million subscribers reported in September 2021. The platform's advertising revenues further underscore its commercial success, nearing eight billion U.S. dollars during the third quarter of 2023 [2].

Moreover, YouTube's global viewership has surpassed 2.5 billion as of January 2024, securing its position as the second most popular social network worldwide by the number of monthly active users [1]. The platform's advertising revenue alone exceeded 31.5 billion U.S. dollars in 2023, demonstrating its unparalleled influence within the digital advertising ecosystem [3].

Given these developments, YouTube's role as a predominant video platform is undeniable, warranting in-depth academic and professional scrutiny. Its profound impact on global video consumption trends, coupled with its significant economic contributions, positions YouTube as a central subject of study within the evolving digital landscape.

Continuous research on video popularity analysis has led to the evolution of predictive models over the past decades. In this paper, we present a novel prediction model that leverages state-of-the-art action detection techniques. Our contributions are twofold.

- **Novel Feature Representations:** We introduce innovative representations of characteristics for video stream data by utilizing emerging video processing algorithms.
- **Dual Architectures:** We design and implement two distinct architectures for the prediction model, one based on the Bag-of-Words framework and the other utilizing BERT embeddings.

These advancements demonstrate significant potential in improving the accuracy and efficiency of video popularity predictions.

2 Previous work

In 2008, Crane and Sornette analyzed patterns in the evolution of YouTube video popularity, identifying four primary categories: "memoryless" videos without a popularity peak; "junk" videos that experience a short-lived burst; "viral" videos with a slow increase, a significant word-of-mouth burst, then a decline; and "quality" videos that peak suddenly due to external events, such as being featured on YouTube's first page, followed by a gradual decrease as they are shared among users [7].

Subsequent research has delved deeply into video popularity prediction, particularly after its predictability was established. Initial studies, such as those by Szabo and Huberman in 2010, focused on the correlation between early and future views. They proposed three models, with the most effective one indicating that future popularity can be predicted from early view counts using a constant factor α determined through linear regression [17].

During this period, the Content Scaling (CS) and Linear Regression (LR) models emerged as significant early prediction methodologies [17]. Meanwhile, Figueiredo et al. contributed by examining

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the evolution of video popularity post-upload and the impact of various referrer types [10]. Figueiredo also introduced K-spectral clustering for predicting popularity based on Crane and Sornette's classes using view count time series [10].

The advancement of prediction models continued with Pinto and colleagues proposing the Multivariate Linear (ML) model and the Multilinear Regression with Radical Bias Function (MRBF) model, the latter becoming widely accepted due to its superiority in handling feature vectors and video similarity assessments [15]. Their MRBF model, which can be implemented as linear or ridge regression, can be written as:

$$NP_i(v, t_r, t_t) = \theta_{(t_r, t_t)}^* \cdot X_{t_r}^*(v) \quad (1)$$

where v denotes the video, t_r represents the number of days after publication used for prediction, and t_t is the target day for prediction. θ^* is the concatenation of weights, and X^* is the number of previous views concatenated with RBF features.

Building on Pinto's frameworks, researchers have subsequently developed various extensions and adaptations of the MRBF model, integrating different features and transitioning from linear to more sophisticated prediction methodologies.

By 2013, Roy et al. introduced a novel approach utilizing social stream data for better predicting videos likely to experience sudden popularity surges[16]. In contrast to the predominant focus on non-visual data, Khosla et al. demonstrated the value of visual content in predicting social media image popularity, an approach further refined by Gelli et al. through the integration of visual sentiment analysis[12, 14].

Visual and sentiment features are also utilized for popularity prediction of online videos by Fontanini et al. [13] Building on the MRBF model, they use multi linear regression as the baseline and incorporate visual and sentimental feature extracted by deep CNNs. Their model is referred to as PMRBFV, where "P" represents popularity trends and "V" denotes visual features. The expression of their model is similar to MRBF:

$$NP_i(v, t_r, t_t) = \theta_{(t_r, t_t)}^* \cdot X_{t_r}^*(v) \quad (2)$$

except that X^* contains BRF features driven from visual and sentiment features from each popularity trend video collection C and the weights for each trend collection are include in θ^* . The visual and sentiment part of PMRBFV model alone is:

$$\sum_{v_c \in C} \omega_{v_c} \cdot RBF_{v_c}(v) \quad (3)$$

Beyond basic multi-linear regression approaches, more advanced machine learning and deep learning methods has been applied to popularity prediction. Trzciński et al. proposed using Support Vector Regression for predicting video popularity[18]. In 2018, a multimodal deep learning method is introduced by Chen et al. for popularity prediction of online videos[6].

3 Our model

To evaluate and compare model performance on predicting video popularity, we use the Relative Squared Error (RSE). Unlike Mean Squared Error (MSE), which is sensitive to outliers, RSE is more appropriate for this task due to the significant variability in video

popularity. This characteristic makes RSE a more robust and meaningful metric in this context. The definition of RSE is given by:

$$RSE = \left(\frac{\hat{N}(v, t_r, t_t)}{N(v, t_t)} - 1 \right)^2 \quad (4)$$

Here, v denotes the video, t_r represents the number of days after publication used for prediction, and t_t is the target day for prediction.

$\hat{N}(v, t_r, t_t)$ is the predicted view count for video v based on view data up to day t_r and visual features of v , and $N(v, t_t)$ is the actual view count on the day t_t .

For a collection of videos C , the performance metric is the arithmetic mean of RSE for all videos in C .

3.1 Feature

3.1.1 Previous views. Following prior approaches [10, 13, 15, 17], we utilize the previous daily view counts of a video from its upload day 0 to day t_r .

Let $x_i(v)$ denote the number of views of video v on the day t_i , the views input is then a vector $X_{t_r} = (x_1(v), x_2(v), \dots, x(t_r))$. Note that $x_0(v)$ is excluded, as the view count on day 0 is always 0.

3.1.2 Visual features. Previous studies have demonstrated that visual features, together with view data can enhance the performance of prediction models[12–14]. Building on this insight, we proposed that improved visual feature representations can further refine predictions by offering a deeper understanding of the factors influencing the popularity of online videos.

To achieve this, we leverage state-of-the-art feature extractors designed to capture chronological visual features of action, enabling more comprehensive video analysis.

The SlowFast model is a deep learning architecture designed for video recognition and video understanding[9]. It is particularly well-suited for action recognition task: the model processes video inputs and label the actions occurring within a specific frame duration. The SlowFast model has two pathways, enables it to effectively understand both fast and slow motions in videos. The slow pathway operates at a low frame rate, capturing the overall spatial semantics and slower movements, while the fast pathway operates at a high frame rate, capturing fast-moving actions.

The output of SlowFast for each video is an array of action labels, with each label corresponding to a specific segment of frames within the video.

For our task, we utilize a pre-trained SlowFast model trained on the Kinetics-400 dataset, which includes a total of 400 action labels. To adapt the model to our requirements, the frame rate is adjusted to align with the specifics of the task.

3.2 Visual feature representation

We have proposed two variants of our method, both of which share the same backbone predictor, a multi-layer perceptron (MLP). However, they differ in how they process and represent visual content.

3.2.1 Variant I: Bag-of-words. As the raw visual content features are extracted in the format of a word array, our first variant leverages the bag-of-words approach to process these features. The architecture of this model is shown in Figure 1.

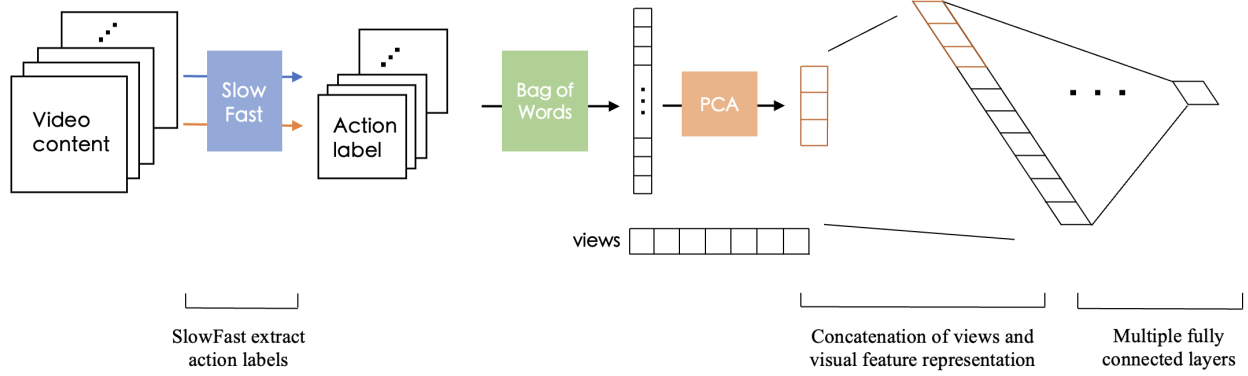


Figure 1: Bag-of-words architecture.

As the extracted content features correspond to a predefined set of action labels, this method counts the frequency of each action label in the video. Using these frequencies, the content features are projected into a new vector space, where each dimension corresponds to an action label, and the scalar value captures its multiplicity. This transformation standardizes the content features into a fixed vector space regardless of their varying shape. Through the bag-of-words transformation, visual content features are uniformly represented as vectors of size 400, corresponding to the total number of action labels.

The resulting features are sparse matrices because most videos contain very few labels. To mitigate sparsity and increase stability, we further refine and compress these features, making them more representative. By applying Principal Component Analysis (PCA), only the most significant dimensions are retained from the feature matrices. Through such transformation and compression, the visual features representations are concise. It will then be concatenated with the view features to serve as input for the neural network predictor.

3.2.2 Variant II: BERT Embedding. In Variant I, we process and compress the representation of visual content using a bag-of-words approach followed by PCA. While these methods effectively capture the presence and multiplicity of actions, they disregard the sequential order and semantical features of actions. The raw visual content consists of a chronological sequence of action labels throughout the video. The order in which events occur and their underlying semantical context significantly influence video popularity.

Recognizing the importance of chronological and semantical information, we propose a second variant that incorporates these factors into the visual feature representation.

We utilize a BERT model (Bidirectional Encoder Representations from Transformers)[8] to transform the sequence of action labels into rich, meaningful feature representations. Figure 2 shows the architecture. The self-attention mechanism in the model captures both chronological order and contextual relationships between

action labels, while its positional embeddings encode the temporal structure of the sequence.

BERT’s contextualized embeddings go beyond surface-level representation by capturing the semantic information of action labels. This includes their individual meanings and how they interact within the sequence, allowing the model to understand nuances such as relationships, dependencies, and implied context.

The input sequence of action labels is tokenized into a fixed-length vector using BERT’s tokenizer, which ensures consistent input size for the BERT encoder. This enables the extraction of high-dimensional embeddings that encode both the semantical and temporal dynamics of the video content, offering a comprehensive representation for downstream prediction tasks.

To obtain sentence-level embeddings for the action label sequence, we take output from the last hidden layer of BERT model. During the encoding process, each token in the input sequence is transformed from a scalar value into a high-dimensional vector of size v (e.g., 568 for BERT-Base). After this transformation, the input matrix of size c_i (where c_i is the sequence length, i.e., the number of tokens) becomes a matrix of size $c_i \times v$, where v represents the output dimension of the last hidden layer.

At this stage, the visual content representation is a two-dimensional matrix, while the view features are a one-dimensional array. To concatenate these features while reducing model complexity, we first compress the visual content representation. We apply average pooling across the embeddings along the feature dimension (v) by calculating the mean of the matrix along this axis. This operation reduces the feature matrix to a one-dimensional vector of size c_i , while preserving the sequence length (c_i).

This approach retains the chronological information embedded in the visual content representation while keep the simplicity of feature dimensions. The resulting representation aligns with the view features, making it suitable for concatenation and further processing in the prediction model.

The input to the prediction model consists of both the view features and the visual content representation. To better process and integrate these two components, we design a fully connected layer

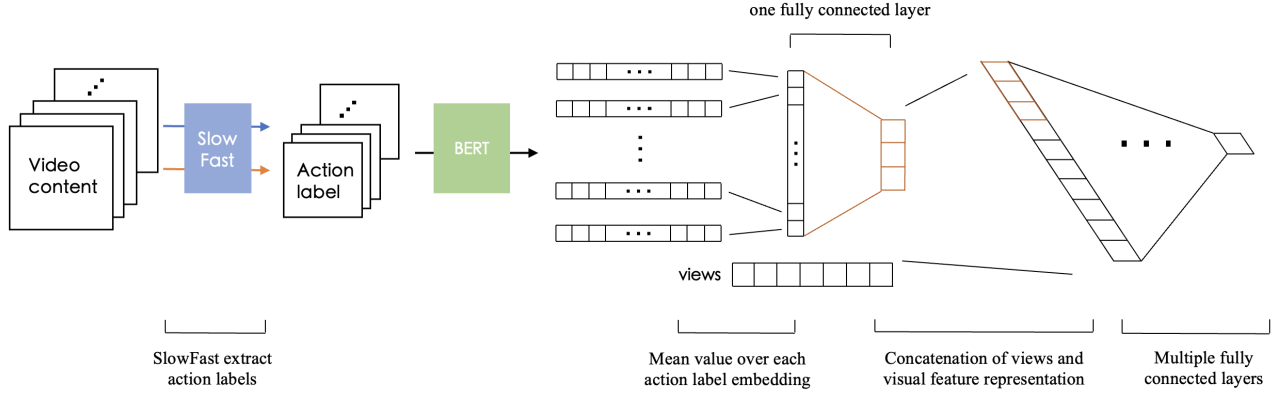


Figure 2: BERT embedding architecture.

header specifically for processing the visual content representation. This header layer processes the visual content representation independently before concatenation with view features.

3.3 Prediction model

Predicting video popularity is formulated as a regression task. Our model employs a deep neural network, specifically a Multilayer Perceptron (MLP) model to perform this prediction. For each video, the model takes two inputs: the daily view counts for the first t_r days after the video's release and the visual content representation.

The input matrix for a single video has a size of $t_r + c$, where t_r represents the number of reference days after the video's release, and c denotes the size of the processed visual content feature representation. The model outputs a scalar value, which corresponds to the predicted view count on the desired day t_t . This architecture integrates temporal and content-based features to improve the accuracy of popularity predictions.

For a MLP with 4 hidden layers, input x , weights $W^{(l)}$, biases $b^{(l)}$, and activation functions $\sigma^{(l)}$, the network can be expressed as follows:

Let $f^{(l)}(x)$ denote the operation of the l -th layer, incorporating its weights, biases, and activation function. The forward pass is given by:

$$y = f^{(5)} \circ f^{(4)} \circ f^{(3)} \circ f^{(2)} \circ f^{(1)}(x), \quad (5)$$

where each hidden layer's function $f^{(l)}$ is defined as:

$$f^{(l)}(z) = \sigma^{(l)}(W^{(l)}z + b^{(l)}). \quad (6)$$

The output activation function $\sigma^{(5)}$ is the identity function, simplifying the final layer to:

$$f^{(5)}(z) = W^{(5)}z + b^{(5)}. \quad (7)$$

4 Experiment

4.1 Dataset

For our task, we use a publicly available dataset on YouTube, originally collected by Figueiredo et al.[11]. This dataset includes the

daily view counts of videos, recorded from the day of their upload (denoted as $t_0 = 0$). It has been widely utilized in previous studies on video popularity prediction, including Pinto et al.[15] and Fontanini et al.[13]

The dataset is divided into two subsets based on the video collection method. The first subset, referred to as the Random dataset, contains videos randomly selected from the entire pool of available videos. The second subset, known as the Top dataset, comprises videos labeled as top-performing by YouTube.

4.1.1 Data filter. Our task involves both video view data and video content, focusing exclusively on the Top dataset. We begin by filtering out videos that are no longer available on YouTube. Next, we clean the data by removing videos with incomplete records (e.g., records that last shorter than target day for prediction). To eliminate the impact of outliers, we exclude videos with view counts below a threshold $y_l \geq 100$ on the target day.

As mentioned earlier, previous study by Crane and Sornette has demonstrated the existence of distinct patterns in YouTube video view growth[7]. The four categories are Viral, Quality, Junk, and Memoryless. These patterns have been shown to significantly enhance popularity prediction performance: training separate models for each pattern subset outperforms training a single general model on the entire dataset. The classification is based on testing for Poisson distribution properties and analyzing the fraction of views during specific periods relative to total views.

We applied the same classification algorithm to the Top dataset, resulting in four subsets corresponding to the identified patterns. Due to an insufficient number of videos for meaningful training and testing, we discarded the Memoryless subset. The final dataset consists of three subsets: Viral, Quality, and Junk. After filtering, the Viral subset contains 270 videos, the Quality subset 571 videos, and the Junk subset 620 videos.

Each subset is treated as a separate dataset during the training and testing process. Individual models are trained on each subset, resulting in three distinct models for each variant of our method.

4.2 Model and training Set up

The MLP structure remains consistent for both variants. The combined features, with a length of 10 per sample, are fed to an MLP comprising four hidden layers. The output is a single scalar representing the prediction. The activation function used as the non-linear layers of the MLP is the sigmoid.

In the training process, the dataset is randomly split into a 3:1 ratio for training and testing. The maximum number of epochs is set to 3000 with batch size of 32, and AdamW is used as the optimizer. Instead of the standard MSE loss, we use the custom mRSE loss discussed in earlier section to better fit our task. The initial learning rate is set to 0.001, with a weight decay of 0.01 to prevent overfitting. We also applied a linear learning rate scheduler.

4.3 Results

In our experiments, we evaluate the performance of our model in comparison to other baseline models, including the MRBF model[15], and the PMRBFV model[13]. We also add the view only MLP model, which serves as the baseline for deep learning models. This model differs from our predictors solely in its input layer, as it excludes visual features. For all experiments, the target day t_t is set to 30.

Table 1, Table 2, and Table 3 present the mRSE performance of our model's two variants: Bag-of-Words and BERT Embedding compared to the baseline MLP without visual features, evaluated on the Viral, Junk, and Memoryless subsets of the Top dataset respectively.

We reports both our variants' performance along with the baseline MLP on whole Top dataset in Table 4. In the comparison, all models are trained and test on whole top dataset rather than individual trend subsets.

Table 5 shows the percentage of mRSE error reduction against the baseline MLP for both our variants on each reference date and the mean reduction of all reference days. The reduction percentage for each date is calculated as the averaged reduction over the three trend subsets and the whole Top dataset on that date.

The accompanying Figure3 provide a comprehensive comparison of both our variants with other baseline models on the Top dataset across 7 reference days. For consistency of comparison, our model variants and view only MLP are trained on the Top dataset as a whole rather than subsets.

Days	View only	BoW	Embedding
1	0.4580	0.2262	0.2522
2	0.3782	0.1881	0.1627
3	0.3479	0.1712	0.1475
4	0.3134	0.1619	0.1258
5	0.2946	0.1431	0.1188
6	0.2682	0.1121	0.1149
7	0.2277	0.1064	0.0995

Table 1: mRSE performance on Viral subset with $t_t=30$

Days	View only	BoW	Embedding
1	0.3788	0.3048	0.2988
2	0.2834	0.1730	0.1715
3	0.2555	0.1633	0.1501
4	0.2288	0.1420	0.1270
5	0.1952	0.1364	0.1345
6	0.1436	0.1317	0.1193
7	0.1362	0.1169	0.1083

Table 2: mRSE performance on Quality subset with $t_t=30$

Days	View only	BoW	Embedding
1	0.3638	0.3227	0.3259
2	0.2243	0.2167	0.2007
3	0.1779	0.1606	0.1575
4	0.1350	0.1318	0.1306
5	0.1297	0.1298	0.1171
6	0.1295	0.1141	0.1031
7	0.1113	0.1047	0.0966

Table 3: mRSE performance on Junk subset with $t_t=30$

Days	View only	BoW	Embedding
1	0.4102	0.4043	0.4094
2	0.2925	0.2634	0.2552
3	0.2263	0.2098	0.2028
4	0.2059	0.1747	0.1742
5	0.1636	0.1506	0.1580
6	0.1247	0.1206	0.1155
7	0.1183	0.1148	0.1130

Table 4: mRSE performance on Top dataset with $t_t=30$

Days	BoW	Embedding
1	20.7%	19.2%
2	25.6%	29.9%
3	26.0%	30.2%
4	26.0%	30.8%
5	22.4%	26.0%
6	20.4%	25.5%
7	19.1%	23.6%
Mean	22.9%	26.5%

Table 5: average mRSE reduction overview only MLP

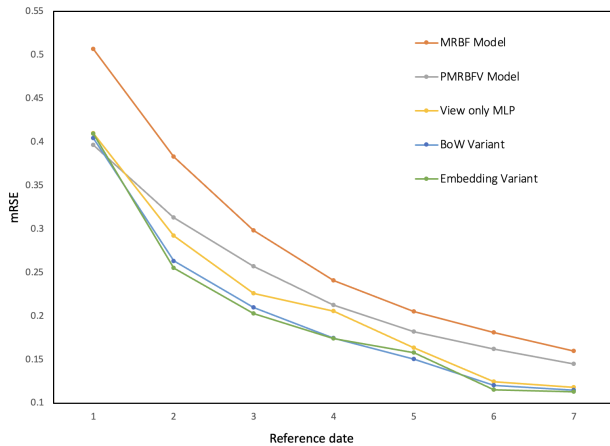


Figure 3: Comparison of methods performance on Top dataset

4.4 Discussion

As Table 1 to 5 has shown, on each subset of Top datasets, both our proposed methods variants outperform the baseline MLP on view features only across all 7 reference days. As anticipated, the additional visual feature improve the prediction result.

More specifically, the improvement is most evident between reference day 2 to 4, where both variants are able to reduce over 25% mRSE error against baseline.

Comparing models' performance between trend sub dataset and whole Top dataset reveals that in general, both our variants benefit from the division of trend based subset. The mRSE error on the whole Top dataset is larger than each individual subset for both variants, which becomes more evident with less reference days.

For the comparison between the two variants, BERT-embedding-based variants achieved better performance over Bag-of-words-based methods overall. It reduce an average of 3.6% mRSE on baseline MLP compared to Bag-of-words variants over all 7 reference days.

The full comparison of methods on whole Top dataset as Figure 3 presents, the deep learning methods achieve better performance than traditional methods from reference day 2 to day 7. In addition, both of our two variants showed improvements on baseline MLP on day 2 to 4, which is consistent to the observation from Table 5.

5 Conclusion

In this paper, we propose utilizing high-level visual features with different representations to predict the popularity of online video content, measured by the number of views. Our method was evaluated on the YouTube Top dataset and across three trend subsets. The results demonstrate its superiority compared to state-of-the-art approaches. The findings indicate that incorporating visual features from videos can significantly enhance the prediction of future video popularity, and further leveraging semantic and chronological information improves the results even more.

However, the dataset used in this study was collected and recorded several years ago, with many videos now unavailable, limiting the

dataset's size. Additionally, YouTube's global viewership has grown substantially over time. For future work, we plan to evaluate our methods on a larger and more up-to-date dataset. We also intend to extend the feature set for videos by incorporating additional semantic features extracted using state-of-the-art video analyzers to gain deeper insights into the factors influencing video popularity.

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